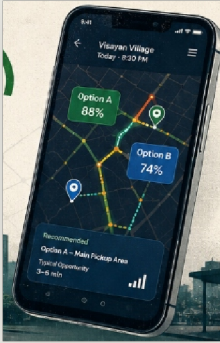


ROUTE SENSE

Know *where* rides are.
Know *when* they're available.



- SMARTER DECISIONS**
Find the best places and times to ride.
- REAL-TIME INSIGHTS**
Up-to-date community intelligence.
- CONNECTED JOURNEYS**
Know your ride. Know your signal.
- PRIVACY FIRST**
Your data stays protected.

1 THE PROBLEM

Knowing a transport route does not mean knowing whether a ride is actually available.
Informal public transportation can change depending on the time, location, demand, traffic, weather, and day of the week. Commuters may already know where vehicles usually pass, but they still face questions such as:

- "Can I actually get a ride here right now?"
- "Where is the better place to wait?"
- "How reliable is transportation at this hour?"
- "Will I still have mobile signal during the trip?"

Existing static route maps cannot fully answer these changing, real-world conditions.

2 BENEFICIARIES

PRIMARY USERS

- Students and daily commuters
- Workers and late-night commuters
- Delivery and gig-economy workers

SECONDARY BENEFICIARIES

- Schools and universities
- Transport cooperatives and associations
- Local government and transport planners
- Researchers
- Telecommunications and mobility planners

3 THE SOLUTION

RouteSense is an Android-based mobility intelligence platform that uses sensors inside passengers' smartphones to understand how informal transportation behaves over time.

- StopSense**
Learns patterns related to passenger movement, pickup areas, and service availability.
- SignalTrail**
Measures cellular connectivity to identify frequently used transport corridors.

- Instead of simply showing commuters where a route exists, RouteSense also provides:
- Where rides are likely available
 - When they are available
 - How reliable the estimate is
 - How connected the journey is

4 HOW IT WORKS

- Opt In**
The commuter voluntarily enables passive mobility sensing.
- Sense Movement**
The phone detects activity changes such as walking, waiting, and travelling in a vehicle.
- Collect Context**
GPS/GNSS, accelerometer, gyroscope, time, and cellular signal measurements provide mobility context.
- Analyze On-Device**
RouteSense estimates probable boarding areas, travel corridors, movement patterns, and connectivity conditions.
- Protect Privacy**
Detailed raw movement data is minimized and converted into anonymous or aggregated observations.
- Build Community Intelligence**
Observations from many trips reveal service patterns, pickup activity, reliability, and connectivity.
- Recommend**
RouteSense suggests transportation options based on current and historical conditions.

5 KEY FEATURES

- 88% Service Probability**
Estimates how likely a commuter is to find transportation at a particular location and time.
- Smart Pickup Areas**
Identifies locations where successful transport activity frequently occurs.
- Opportunity Window**
Shows a realistic waiting range such as **Typical ride opportunity: 4-7 minutes** instead of pretending to know an exact vehicle arrival time.
- Time-Based Mobility**
Learns how transport availability changes between morning, afternoon, evening, weekdays, and weekends.
- Confidence & Freshness**
Every recommendation shows how much recent evidence supports it.

High Confidence • 31 observations at similar times • Last observation: 8 minutes ago
- Connectivity-Aware Routes**
Compares transportation choices based on both mobility and cellular connectivity.
- Route Pattern Discovery**
Detects recurring and emerging transport corridors from aggregated passenger movement.
- Offline Cached Insights**
Previously learned mobility information remains accessible during weak connectivity.
- No Driver App Required**
RouteSense learns from participating passenger smartphones rather than requiring every vehicle to join the platform.

6 NO dedicated GPS tracker or driver application is required.

- Smartphone Sensor Fusion:** Combines multiple phone sensors for movement understanding beyond GPS alone.
- Activity Recognition:** Identifies transitions: *Walking* → *Waiting* → *In Vehicle* → *Walking*.
- GNSS / Geospatial Analytics:** Analyzes movement trajectories and spatial patterns.
- Cellular Signal Intelligence:** Measures network conditions during real passenger journeys.
- Context-Aware Computing:** Considers location, time, activity, and environment when interpreting mobility.
- On-Device Inference:** Processes sensitive movement data locally before sharing minimized insights.
- Geospatial Clustering:** Finds recurring pickup areas, transport corridors, and mobility hotspots.
- Predictive Analytics:** Estimates service probability based on historical and recent observations.
- Privacy-Preserving Aggregation:** Builds community mobility intelligence without publishing individual travel histories.

PHONE HARDWARE USED

- GPS / GNSS
- Accelerometer
- Gyroscope
- Cellular Modem
- Compass / Magnetometer

8 EXAMPLE

Going to Visayan Village				
● OPTION A	Main Pickup Area		● OPTION B	Alternate
	Service Probability	Typical Opportunity	Service Probability	Typical Opportunity
	88%	3-6 min	74%	6-10 min
	✓ Confidence: High	📶 Connectivity: Moderate	✓ Confidence: Moderate	📶 Connectivity: Low
FASTEST CHOICE: OPTION A			BEST CONNECTED: OPTION B	

7 TECH STACK

- Mobile App: React Native + TypeScript
- Android Sensor Layer: Kotlin
- Local Storage: SQLite
- Maps: MapLibre + OSM
- Geospatial Backend: Supabase + PostgreSQL
- Geospatial Processing: PostGIS + H3
- Data Analytics: Python + pandas + scikit-learn
- Predictive Models: XGBoost / lightgbm

9 PROJECT TEAM

- Ryan Christian Deniega
- Kyndel Roy Suarez
- Domince Aseberos
- Anjelo Vidal

CCE 106 — Application Development & Emerging Technologies | Student Researchers

10 REFERENCES & CITATIONS

- [1] City Government of Tagum. City Ordinance No. 984, S. 2021 — Local Public Transport Route Plan.
- [2] Android Developers. Activity Recognition Transition API & GnsMeasurement API.
- [3] Android Developers. TelephonyCallback.SignalStrengthsListener API.
- [4] Behrens et al. (2025). Spatial & temporal variability of informal transit using GPS observations.
- [5] Travel Behaviour & Society (2025). Passenger smartphone GPS to identify travel, waiting, & transfer behavior.